

Foundation (Jalapeno)

Q1. What is the simplified form of $\sqrt{16}$?

- (A) 2
- (B) 4
- (C) 6
- (D) 8
- (E) 10

Q2. The distance between two planets is $\sqrt{64}$ kilometers. How many kilometers is this?

- (A) 4
- (B) 6
- (C) 8
- (D) 10
- (E) 12

Q3. A spacecraft travels at a speed of $\sqrt{81}$ km/h. What is its speed in km/h?

- (A) 6
- (B) 8
- (C) 9
- (D) 12
- (E) 15

Q4. What is the value of $\sqrt[3]{27}$?

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q5. A scientist measures the volume of a cube to be $\sqrt[3]{64}$ cubic meters. What is the volume of the cube?

- (A) 2
- (B) 4
- (C) 6
- (D) 8
- (E) 10

Q6. The equation $x^2 = 16$ can be written as $x = \pm\sqrt{16}$. What is the value of x ?

- (A) 2
- (B) 4
- (C) ± 2
- (D) ± 4
- (E) ± 6

Q7. What is the rational exponent form of $\sqrt[3]{x}$?

- (A) $x^{1/2}$
- (B) $x^{1/3}$
- (C) $x^{2/3}$
- (D) $x^{3/4}$
- (E) $x^{4/5}$

Q8. The equation $x^{1/2} = 4$ can be written as $x =$

- (A) $\sqrt{4}$
- (B) $\sqrt[3]{4}$
- (C) 4^2
- (D) 4^3
- (E) 16

Open Ended Questions (FRQ)

Q9. A lab experiment involves measuring the distance of a projectile launched from a spacecraft. The distance traveled is given by $d = \sqrt{t^2 + 4t + 4}$. Simplify the expression and find the distance traveled when $t = 2$ seconds.

Q10. The volume of a spherical planet is given by $V = \frac{4}{3}\pi r^3$. Convert the equation to rational exponent form and find the volume when $r = 4$ kilometers.

Chef's Tips

- Ensure students use the correct notation for radicals and rational exponents.
- Remind students to simplify expressions with radicals whenever possible.
- Encourage students to check their work by plugging their answers back into the original equation.